STATS I · WEEK 1

Stats I week 1 formulas: Types of data and scales

Data, population & sample · Variable types & scales of measurement. Save this PDF for revision.

DATA, POPULATION & SAMPLE · IN EVERY PAPER

Data, population & sample

Term	Meaning
Population	The entire group you want to study (all voters in India)
Sample	The subset you actually collect data from
Parameter	A number describing the population (μ, σ)
Statistic	A number computed from a sample ($\bar{x}$, s)
Descriptive statistics	Organising and summarising data: tables, charts, mean, SD

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Data, population & sample

Term	Meaning
Inferential statistics	Using a sample to draw conclusions about the population
Cross-sectional data	Many units at one point in time (marks of 500 students in one exam)
Time-series data	One unit tracked over time (a company's yearly profit, 2015–2024)
Case / observation	A row of the data table (one unit)
Variable	A column: a characteristic recorded for every case

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Key points

- For inference to be valid, the sample must be representative of the population: randomly chosen and unbiased.
- A claim about every organisation in a city based on one organisation's data is an over-generalisation: the sample doesn't represent the population.

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Common traps

⚠ Data from many villages or students collected at one time is cross-sectional, even if the table has a 'year' column.

⚠ Survey 100 students to learn about all IITM students: population = all IITM students, sample = the 100.

VARIABLE TYPES & SCALES OF MEASUREMENT · IN EVERY PAPER

Categorical vs numerical

Type	What it is	Examples
Categorical (qualitative)	Labels or groups	Gender, city, blood group, T-shirt size, PIN code, roll number
Numerical, discrete	Countable values with gaps	Children in a family, cars sold, goals scored
Numerical, continuous	Any value in a range (measured)	Height, weight, time, temperature

VARIABLE TYPES & SCALES OF MEASUREMENT · IN EVERY PAPER

Scales: N-O-I-R, each adds one property

Scale	Adds	Allowed	Examples	Centre
Nominal	Names only, no order	= ≠	Gender, religion, election symbol, city, jersey number	Mode
Ordinal	Order (gaps unequal)	< >	Ratings, ranks, grades, sizes S/M/L/XL, education level	Mode, median
Interval	Equal gaps, no true zero	+ −	Temperature in °C/°F, calendar year, time of day	Mode, median, mean
Ratio	True zero (0 = none)	+ − × ÷	Height, weight, income, age, sales, number of votes	All

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Common traps

⚠ Tip: Ask in order: Is it just a label? → nominal. Can I put it in order? → ordinal. Does 0 mean 'none of it'? → ratio; if not → interval.

⚠ Numbers used as labels (PIN code, roll number, phone number, jersey number) are categorical, nominal.

⚠ °C is interval: 0 °C isn't 'no temperature', and 20 °C isn't twice as hot as 10 °C.

⚠ Counts (votes, children, cars) are numerical, discrete, ratio scale.

⚠ PYQ (May 2024): election symbol is categorical with a nominal scale.

Notes for every week of Stats I, free.

Full notes, the formula sheet and common traps, with worked PYQs.

btechi.in/qualifier/notes/stats-1 →

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